Homework 03 solution

1.

IDENTIFY: In each case the forces are constant and the displacement is along a straight line, so $W = Fs\cos\phi$.

SET UP: In part **(a)**, when the cable pulls horizontally $\phi = 0°$ and when it pulls at $35.0°$ above the horizontal $\phi = 35.0°$. In part **(b)**, if the cable pulls horizontally $\phi = 180°$. If the cable pulls on the car at $35.0°$ above the horizontal it pulls on the truck at $35.0°$ below the horizontal and ϕ 145.0°. For the gravity force $\phi = 90°$, since the force is vertical and the displacement is horizontal.

EXECUTE: **(a)** When the cable is horizontal, $W = (850\text{ N})(5.00\times10^3\text{ m})\cos 0° = 4.26\times10^6\text{ J}$. When the cable is $35.0°$ above the horizontal, $W = (850\text{ N})(5.00\times10^3\text{ m})\cos 35.0° = 3.48\times10^6\text{ J}$.

(b) $\cos 180° = -\cos 0°$ and $\cos 145.0° = -\cos 35.0°$, so the answers are $-4.25\times10^6\text{ J}$ and $-3.48\times10^6\text{ J}$.

(c) Since $\cos\phi = \cos 90° = 0$, $W = 0$ in both cases.

EVALUATE: If the car and truck are taken together as the system, the tension in the cable does no net work.

2.

IDENTIFY: From the work-energy relation, $W = W_{\text{grav}} = \Delta K_{\text{rock}}$.

SET UP: As the rock rises, the gravitational force, $F = mg$, does work on the rock. Since this force acts in the direction opposite to the motion and displacement, *s*, the work is negative. Let *h* be the vertical distance the rock travels.

EXECUTE: **(a)** Applying $W_{\text{grav}} = K_2 - K_1$ we obtain $-mgh = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$. Dividing by *m* and solving for v_1, $v_1 = \sqrt{v_2^2 + 2gh}$. Substituting $h = 15.0$ m and $v_2 = 25.0$ m/s,

$$v_1 = \sqrt{(25.0\text{ m/s})^2 + 2(9.80\text{ m/s}^2)(15.0\text{ m})} = 30.3\text{ m/s}$$

(b) Solve the same work-energy relation for *h*. At the maximum height $v_2 = 0$.

$$-mgh = \tfrac{1}{2}mv_2^2 - \tfrac{1}{2}mv_1^2 \text{ and } h = \frac{v_1^2 - v_2^2}{2g} = \frac{(30.3\text{ m/s})^2 - (0.0\text{ m/s})^2}{2(9.80\text{ m/s}^2)} = 46.8\text{ m}.$$

EVALUATE: Note that the weight of 20 N was never used in the calculations because both gravitational potential and kinetic energy are proportional to mass, *m*. Thus any object, that attains 25.0 m/s at a height of 15.0 m, must have an initial velocity of 30.3 m/s. As the rock moves upward gravity does negative work and this reduces the kinetic energy of the rock.

3.

IDENTIFY: Apply the relation between energy and power.

SET UP: Use $P = \frac{W}{\Delta t}$ to solve for W, the energy the bulb uses. Then set this value equal to $\frac{1}{2}mv^2$ and solve for the speed.

EXECUTE: $W = P\Delta t = (100\text{ W})(3600\text{ s}) = 3.6\times10^5\text{ J}$

$K = 3.6\times10^5\text{ J}$ so $v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2(3.6\times10^5\text{ J})}{70\text{ kg}}} = 100\text{ m/s}$

EVALUATE: Olympic runners achieve speeds up to approximately 10 m/s, or roughly one-tenth the result calculated.

4.

IDENTIFY: $U_{\text{grav}} = mgy$ so $\Delta U_{\text{grav}} = mg(y_2 - y_1)$

SET UP: $+y$ is upward.

EXECUTE: **(a)** $\Delta U = (75\text{ kg})(9.80\text{ m/s}^2)(2400\text{ m} - 1500\text{ m}) = +6.6\times10^5\text{ J}$

(b) $\Delta U = (75\text{ kg})(9.80\text{ m/s}^2)(1350\text{ m} - 2400\text{ m}) = -7.7\times10^5\text{ J}$

EVALUATE: U_{grav} increases when the altitude of the object increases.

5.

IDENTIFY: Use the information given in the problem with $F = kx$ to find k. Then $U_{\text{el}} = \frac{1}{2}kx^2$.

SET UP: x is the amount the spring is stretched. When the weight is hung from the spring, $F = mg$.

EXECUTE: $k = \frac{F}{x} = \frac{mg}{x} = \frac{(3.15\text{ kg})(9.80\text{ m/s}^2)}{0.1340\text{ m} - 0.1200\text{ m}} = 2205\text{ N/m}.$

$x = \pm\sqrt{\frac{2U_{\text{el}}}{k}} = \pm\sqrt{\frac{2(10.0\text{ J})}{2205\text{ N/m}}} = \pm0.0952\text{ m} = \pm9.52\text{ cm}.$ The spring could be either stretched 9.52 cm compressed 9.52 cm. If it were stretched, the total length of the spring would be $12.00\text{ cm} + 9.52\text{ cm} = 21.52\text{ cm}$. If it were compressed, the total length of the spring would be $12.00\text{ cm} - 9.52\text{ cm} = 2.48\text{ cm}$.

EVALUATE: To stretch or compress the spring 9.52 cm requires a force $F = kx = 210\text{ N}$.

6.

IDENTIFY: Apply $W_{f_k} = f_k s\cos\phi$. $f_k = \mu_k n$.

SET UP: For a circular trip the distance traveled is $d = 2\pi r$. At each point in the motion the friction force and the displacement are in opposite directions and $\phi = 180°$. Therefore, $W_{f_k} = -f_k d = -f_k(2\pi r)$. $n = mg$ so $f_k = \mu_k mg$.

EXECUTE: **(a)** $W_{f_k} = -\mu_k mg2\pi r = -(0.250)(10.0\text{ kg})(9.80\text{ m/s}^2)(2\pi)(2.00\text{ m}) = -308\text{ J}$.

(b) The distance along the path doubles so the work done doubles and becomes -616 J.

(c) The work done for a round trip displacement is not zero and friction is a nonconservative force.

EVALUATE: The direction of the friction force depends on the direction of motion of the object and that is why friction is a nonconservative force.

7.

IDENTIFY and **SET UP:** $p = mv$. $K = \frac{1}{2}mv^2$.

EXECUTE: (a) $p = (10{,}000\text{ kg})(12.0\text{ m/s}) = 1.20\times10^5\text{ kg}\cdot\text{m/s}$

(b) (i) $v = \dfrac{p}{m} = \dfrac{1.20\times10^5\text{ kg}\cdot\text{m/s}}{2000\text{ kg}} = 60.0\text{ m/s}$. (ii) $\frac{1}{2}m_T v_T^2 = \frac{1}{2}m_{SUV}v_{SUV}^2$, so

$$v_{SUV} = \sqrt{\frac{m_T}{m_{SUV}}}v_T = \sqrt{\frac{10{,}000\text{ kg}}{2000\text{ kg}}}(12.0\text{ m/s}) = 26.8\text{ m/s}$$

EVALUATE: The SUV must have less speed to have the same kinetic energy as the truck than to have the same momentum as the truck.

8.

IDENTIFY: For each object, $\vec{p} = m\vec{v}$ and $K = \frac{1}{2}mv^2$. The total momentum is the vector sum of the momenta of each object. The total kinetic energy is the scalar sum of the kinetic energies of each object.
SET UP: Let object A be the 110 kg lineman and object B the 125 kg lineman. Let $+x$ be to the right, so $v_{Ax} = +2.75$ m/s and $v_{Bx} = -2.60$ m/s.
EXECUTE: **(a)** $P_x = m_A v_{Ax} + m_B v_{Bx} = (110\text{ kg})(2.75\text{ m/s}) + (125\text{ kg})(-2.60\text{ m/s}) = -22.5\text{ kg}\cdot\text{m/s}$. The net momentum has magnitude $22.5\text{ kg}\cdot\text{m/s}$ and is directed to the left.

(b) $K = \frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2 = \frac{1}{2}(110\text{ kg})(2.75\text{ m/s})^2 + \frac{1}{2}(125\text{ kg})(2.60\text{ m/s})^2 = 838\text{ J}$

EVALUATE: The kinetic energy of an object is a scalar and is never negative. It depends only on the magnitude of the velocity of the object, not on its direction. The momentum of an object is a vector and has both magnitude and direction. When two objects are in motion, their total kinetic energy is greater than the kinetic energy of either one. But if they are moving in opposite directions, the net momentum of the system has a smaller magnitude than the magnitude of the momentum of either object.

9.

IDENTIFY: Apply conservation of momentum to the system of the astronaut and tool.
SET UP: Let A be the astronaut and B be the tool. Let $+x$ be the direction in which she throws the tool, so $v_{B2x} = +3.20$ m/s. Assume she is initially at rest, so $v_{A1x} = v_{B1x} = 0$. Solve for v_{A2x}.
EXECUTE: $P_{1x} = P_{2x}$. $P_{1x} = m_A v_{A1x} + m_B v_{B1x} = 0$. $P_{2x} = m_A v_{A2x} + m_B v_{B2x} = 0$ and

$v_{A2x} = -\dfrac{m_B v_{A2x}}{m_A} = -\dfrac{(2.25\text{ kg})(3.20\text{ m/s})}{68.5\text{ kg}} = -0.105$ m/s. Her speed is 0.105 m/s and she moves opposite to the direction in which she throws the tool.
EVALUATE: Her mass is much larger than that of the tool, so to have the same magnitude of momentum as the tool her speed is much less.

10.

8.24. **IDENTIFY:** In part (a) no horizontal force implies P_x is constant. In part (b) use the energy expression, Eq. 7.14, to find the potential energy initially in the spring.
SET UP: Initially both blocks are at rest.

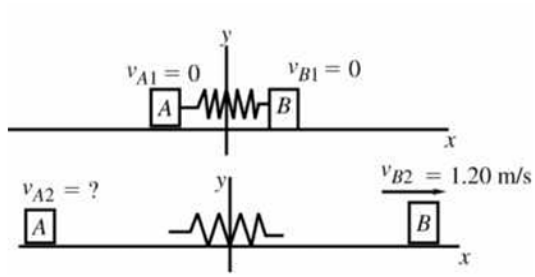

Figure 8.24

EXECUTE: **(a)** $m_A v_{A1x} + m_B v_{B1x} = m_A v_{A2x} + m_B v_{B2x}$

$$0 = m_A v_{A2x} + m_B v_{B2x}$$

$$v_{A2x} = -\left(\frac{m_B}{m_A}\right) v_{B2x} = -\left(\frac{3.00\text{ kg}}{1.00\text{ kg}}\right)(+1.20\text{ m/s}) = -3.60\text{ m/s}$$

Block A has a final speed of 3.60 m/s, and moves off in the opposite direction to B.
(b) Use energy conservation: $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$.
Only the spring force does work so $W_{\text{other}} = 0$ and $U = U_{\text{el}}$.
$K_1 = 0$ (the blocks initially are at rest)
$U_2 = 0$ (no potential energy is left in the spring)

$$K_2 = \tfrac{1}{2} m_A v_{A2}^2 + \tfrac{1}{2} m_B v_{B2}^2 = \tfrac{1}{2}(1.00\text{ kg})(3.60\text{ m/s})^2 + \tfrac{1}{2}(3.00\text{ kg})(1.20\text{ m/s})^2 = 8.64\text{ J}$$

$U_1 = U_{1,\text{el}}$ the potential energy stored in the compressed spring.
Thus $U_{1,\text{el}} = K_2 = 8.64$ J

EVALUATE: The blocks have equal and opposite momenta as they move apart, since the total momentum is zero. The kinetic energy of each block is positive and doesn't depend on the direction of the block's velocity, just on its magnitude.